CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in building AI products from design to implementation and iteration, from LLM-based assistants and NLP pipelines to machine learning and deep learning models in PyTorch and TensorFlow/Keras, measured against ground truth. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by data science work, statistical modelling, and Python, SQL and model deployment experience.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Built and owned an AI product end-to-end as sole developer: an LLM-based knowledge-extraction system for the German energy sector, from design and data ingestion to implementation, deployment and iteration, reporting directly to the CTO and working with SAP domain experts.
- Built a multi-agent NLP system in LangGraph (entity-extraction, relationship-analysis, consolidation and reflection agents in parallel) and measured it against an SAP-system-derived ground truth, raising precision/recall from 33%/45% to 93%/90%.
- Built a deterministic, code-based pipeline instead of an LLM for a second, high-volume data stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing accuracy toward 100%.
- Designed a Model Context Protocol (MCP) service that exposes the knowledge graph to AI agents with millisecond-latency retrieval, integrated into Claude Code.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) on Azure where domain experts validate and approve knowledge-graph changes.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, MCP, RAG, GraphRAG, Vector Search, FastAPI, Postgres, MongoDB, Neo4j, Docker, Kubernetes, Azure, React**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Advised business stakeholders on AI automation: mapped their decision-making workflows, identified where agentic AI could support them and translated their requirements into a solution approach.
- Built and deployed an agentic AI application with LangGraph that routes requests across vector-database retrieval, general inference and summarization.
- Built the Python/FastAPI backend (APIs tested with Postman) and a React frontend, containerized with Docker and deployed to AWS with CDK; version control in GitLab with CI/CD.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD, Postman, React**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated evaluation of RAG retrieval and generation with RAGAS and PytrecEval against chemist-provided ground truth, running as an evaluation pipeline on AWS (Lambda, SQS, DynamoDB) built with CDK for PrismaChat.
- Benchmarked multilingual embedding models on over 1M chunks of internal chemical data and designed RAG pipelines (chunking strategies, vector databases, indexing, fine-tuning models for retrieval).
- **Tech stack: Python, RAGAS, PytrecEval, LangChain, Hugging Face Transformers, PyTorch, AWS (Lambda, SQS, DynamoDB, CDK), Azure AI Search, Docker**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM (TensorFlow/Keras) models to predict prepolymer synthesis outcomes from time-series process data, with engineered lag features and rolling statistical windows.
- Benchmarked the models against empirical curve-fitting, improving early identification of batches likely to deviate from specification, and delivered a Streamlit dashboard for chemists to simulate parameter changes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, SciPy, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet and LSTM models for EV load time-series forecasting; the LSTM reached 96% test accuracy with strong performance on unseen data.

PROJECTS

Project: Meme Generation using LLMs | Open Llama 3B | QLoRA Fine-Tuning
- Fine-tuned an Open Llama 3B model with QLoRA, which accelerated training, to generate responses to educational meme-related prompts.

Project: Enhancing Image Classification | CNN | RNN | ResNet
- Applied CNN, RNN and ResNet deep learning architectures for multi-dataset image classification on (permuted) MNIST and Fashion-MNIST, achieving 93% accuracy.

Project: Analysis of Life Expectancy Data | ML | Scikit-Learn
- Statistical analysis of life expectancy data with regression models, data visualization and feature selection.

TECHNICAL SKILLSET

- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Keras, Scikit-Learn, XGBoost, time-series models (ARIMA, SARIMA, Prophet, LSTM), regression, feature selection
- **LLMs & NLP:** LLM-based assistants and agents (LangGraph, LangChain, LiteLLM), RAG, Hugging Face Transformers, NLTK, QLoRA fine-tuning, evaluation (RAGAS, PytrecEval, precision/recall)
- **Programming & Model Deployment:** Python, SQL, FastAPI, version control (Git, GitLab CI/CD), Docker, Kubernetes, AWS (CDK, Lambda, EC2), Azure
- **Data:** Pandas, NumPy, SciPy, Matplotlib, Postgres, MongoDB, Neo4j, vector databases (Qdrant, Pinecone, Azure AI Search)
- **Ways of working:** cross-functional collaboration with domain experts and stakeholders, AI-assisted development with Claude Code, Agile

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering — August 2017 – July 2021
Minor in Computer Science & Engineering
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - The Complete Neural Networks Boot camp
- Udemy - LLMs Mastery : Transformers & Generative AI

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)